
UNIT 2 APPLICATION OF COMPUTERS

Structure

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2.0 OBJECTIVES

After studying this unit, you should be able to:

- highlight the role of computers for the organization and society;
- highlight the significance of computers in online business, trade and commerce;
- highlight the usage of computers in online banking and finance; and
- highlight the importance of computer networks.

2.1 INTRODUCTION

In the previous unit, we learnt how computer have changed in size, shape, usage etc. over period of time. This particular unit will throw a light on how the computers can be applied in daily life to serve the purpose in totality. As the 21st century is looming ahead, we are witnessing the advancement that humanity may never have dreamt of and one of these developments surely is “Computer”. Computers have left their footprints in every area. It can do all the functions at a speedy rate and also helps searching and progressing in every arena be it our homes or businesses applications and usage of computers can be seen everywhere. In coming days, computers will be used to achieve the brilliant goals of success and incomparable development. In India, the accurate knowledge and use of computers will bring an immense change in astonishing way. Computers have their utility everywhere in every

field they are rapidly being used to the work at a great speed and accuracy where data is so complicated to cope up with by the human brains. Computers make such things easy for us; they can store huge data about inventory, plans, procedures etc. With internet, computers are very powerful tools which can be used to share your knowledge and ideas to help others learn or to get guidance from the experts. In short, applications or uses of a computer are countless but basically, they can be described only in three operations namely Process, Store, Transfer.

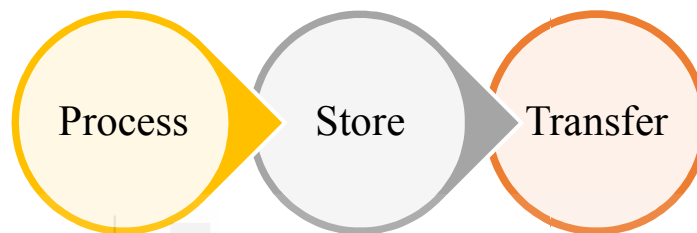


Fig. 2.1: Computer Applications Description

Let us understand this with some examples, by using spreadsheets computers can processes any financial data and organize it into tables, charts, graphs and reports etc. which can be stored on drive. And the stored files can be transferred to any other person or entity via email. While working on an operating system lets say, Windows 10. You install (store) Windows 10 on your hard drive. When you run a computer, Windows 10 loads (transfers) from hard drive to RAM. The services of Windows 10 are run (processed) by CPU. Let's take another example of a YouTube video. You download (transfer from YouTube server to your computer) a YouTube video and save (store) it on your hard drive. Upon double clicking it, it may load (transfer) into RAM and play (process) in VLC media player. Computers have their significance in every arena be it organization, society, business trade and commerce, online business, online banking and finance that is explained in the further sections of the unit.

2.2 ROLE OF COMPUTERS IN BUSINESS ORGANISATION

Computers have brought away organisations a long way ahead from the age of pen, papers and folders stored in dusty storage of compartments. There are various usages of computers in business organizations from generating reports to communicating with clients. For organizations the role of computer is to support the key aspects of running them, such as communication, record-keeping, decision making, data analysis and more. Every aspect of the modern workplace is improved with the help of computers. They are used in every aspect of a company's operations, including product creation,

marketing, accounting and administration. Every aspect of the modern workplace is improved by the use of computers. Computers have made possible the concepts such as flexible working schedules and remote work forces have become possible enabling employees to work from wherever they want and at whatever times they wish. Some of the applications of computers in business organizations are explained below:

- 1) **Communication:** Computers play a very important role in business organisations to maintain a smooth flow of information and communication between different parties and stakeholders. Business can use various methods of communication such as email, live chat tools, video or phone conferencing, social networking, web conferencing etc. It saves times and also improves the functioning of the organisation. With such advance tools, it also becomes easy for the business to offer customer support to its clients in a timely and efficient manner.
- 2) **Marketing:** Organizations can widely use computers for marketing as most of the marketing is done online via social media, blogging, website development, advertising, e-newsletter etc. IT developers can professionally create a website by using various enticing graphics and content with different forms of media, such as text, images, and videos to make the campaign more attractive and lucid for the audience. By using Search Engine Optimization (SEO) and Web Analytics, the website would appear prominently in Google's search results, which then attracts traffic and ultimately enables the business to sell those products to website visitors.
- 3) **Accounting:** Computers can also help organizations in sorting its large financial data in order to obtain their large and small scale financial positions just by few clicks. Computers are also useful for performing various tasks like invoicing clients, maintaining data about debtors and creditors, calculating payroll, calculating and filing tax forms etc. With the computer, accountants can now focus on the higher level picture of a business's finances, leaving the lower level stuff to the machines.
- 4) **Storage:** Businesses can store enormous data on computers. They don't need huge cabinets for storing data now. With computers and server, organizations are capable of storing and sorting billions of files and enable them to access at any time. Digital storage is much larger and more efficient and secure while compared with the physical storage, as it is more difficult to lose or steal files from a well protected database which is one of the main benefits of the computer.
- 5) **Educational Purposes:** With the access of internet on the computers, organizations can reach the pool of world's information which can be used efficiently to educate its employees such as about the company

policies, software use, as well as standard procedure and safety. Instead of hiring teachers' employees can be educated online. Live webinar, conferences and question and answer sessions can also be conducted. Employees can learn through various media, such as videos, text, etc. which enables them to better understand the subject matter.

- 6) **Research:** Computers make business organisations capable of doing a variety of functions, including research about the competition, as well as knowing about the taste and preferences of its customers. With modern data-mining techniques, a business can acquire insightful information about its customers and competition from all kinds of places, including forums, search engines, industry-specific websites and even from social networks. By conducting online surveys, business can know about the public opinions or while launching any new products and public reactions can also be known by reviewing customers review on different sites.

2.3 COMPUTERS FOR SOCIETY

Computers have an integral role to play in our daily lives. With the help of computers tedious functions can be performed automatically and more efficiently leading to the savings of many hours which would otherwise have been spent on doing the job manually. Computers with their usages are now penetrating through the modern society. They are rapidly being used throughout the various sectors of society be it airports, restaurants, railway stations, banks, medical, agriculture, ticket booking, payment of bills, insurance and shopping etc. Some of the applications of computers in various sectors are explained below:

- In agriculture, computers can considerably change the way tasks and businesses are carried out all over the world. They are helpful in finding out the best possible kinds of soil, plants and checking out which match of these would result in the perfect crops.
- In health sectors, computers are very capable of bringing along a medical revolution. They are widely being used for research regarding blood groups, medical histories, etc. that in a way leads to a better diagnosing pattern and makes health care faster and more efficient. The knowledge that computers are providing in this field may lead to better use and purchase of medicinal drugs and thus ensuring better health.
- Computers are also useful in observing and predicting the patterns of traffic which is a grand benefit to all and helps people in saving the hassle of getting stuck for hours in the roadblocks and traffics.

- Computers can also be used to improve administration throughout the world. By providing daily accurate information to the administration departments, computers may change the way decisions are taken across the globe. Keeping all the above-mentioned things in mind, we must accept that if used the right way, computers are a gift of science to mankind.

2.4 ROLE OF COMPUTERS IN BUSINESS, TRADE AND COMMERCE

Computers and other technological devices have surely changed the way the various functions and operations are conducted in business, trade and commerce. Computers have their significance in every type of business almost every type of business now rely on computers to automate their traditional processes. Computers are widely being used by the businesses as a part of their daily operations. Computers have important effect on business operations and help them to reap efficiencies and advantages. Irrespective of the size of enterprise, technology has both tangible and intangible advantages for all the parties involved in terms of saving costs and time spent for conducting business transactions. In business, trade and commerce, computers are helpful in research, production, distribution, marketing, banking, team management, business automation, data storage, employee management and to increase productivity at a lower cost in less time with higher quality. The various advantages of computers in business, trade and commerce are explained below:

- 1) **Inventory Management:** Computers have revolutionized the inventory management as all type of business, trade and commerce with various raw materials suppliers, manufacturers, wholesalers, retails and Business to Business (B2B) providers are dependent on inventory management processes. Technologies ranging from automatic scanners to radio frequency identification chips now allow businesses to track their inventory from the moment a company buys it from a wholesaler to the moment the products leave the premises in the hands of a customer. Matching computer information to inventory storage spaces helps associates pull stock as quickly as possible. Companies can quickly compare inventory when it comes in the door to order sizes on the computer screen. Organized, efficient inventory control helps minimize inventory costs while meeting customer demand.
- 2) **Easier Financial Record Keeping:** Financial record keeping has become easier and more convenient than the traditional methods with the help of computer enabled technologies. To manage their accounting and finance tasks now companies can use various advanced software

programs available which in a way help them in minimizing manual processes, reducing costs and ensuring protection against human error. In fact, companies often use programs that sync accounting with point-of-sale terminals and bookkeeping programs, such that each purchase or sale transaction is automatically captured in an accounting platform.

- 3) **Budgeting and Forecasting:** Computers can widely be used for budgeting, forecasting and purpose by using various software and programs such as spreadsheet excel etc. available in the market. This software can help in doing large financial calculations, estimating future sales and business expansions.
- 4) **Storage:** Computers can also be used in storing huge financial data for business by using internet which was earlier needed to be stored in huge cabinets. Computerized storage saves space and provides more data security. Data can be stored centrally for easy access from multiple computers or stored locally for individual use.
- 5) **Efficiency of Operations:** Computers can largely help business in improving their efficiencies by saving involved costs, time and money and understanding its cash flow needs and preserve precious resources. For example, warehouse inventory technologies can assist to understand how best to manage the storage costs of holding the products. Executives can save time and money by holding meetings over the Internet via Skype, video or telephone conferences, Google meet, Zoom, Microsoft team etc. instead at corporate headquarters.
- 6) **Streamlining Business Communications:** One of the many advantages of a computer in business is that it helps employees to communicate more effectively with each other and with external stakeholders such as customers, partners, investors and prospects. Instead of having to hand-write information, business administration professionals can use applications and programs to facilitate communication. There are a number of communication tools that help businesses to effectively communicate such as, email, instant messaging programs like Skype, team collaboration apps like slack, video conferencing tools such as zoom, project management programs such as teamwork. By using communication technology, businesses can increase the speed at which people communicate and remove physical barriers to communication.
- 7) **Managing Customer and Partner Relationships:** Computers have made it possible for the businesses to work in different geographic locations with the help of technology. Many businesses have partners, suppliers and manufacturers in other parts of the world. If customers or partners have a query for the business, they can reach easily by sending an email or message and within few seconds, they can receive a reply.

Being able to communicate and collaborate with them using computer technology enables organizations to improve product quality, enter new markets, improve customer satisfaction and grow their business.

Check Your Progress A

- 1) How computer in organisations leads to improved communication?

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- 2) How are computers bringing medical revolution?

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- 3) What are the advantages of computers to the society?

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- 4) How can computers ensure more security and safety of confidential information?

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2.5 COMPUTER ROLE IN ONLINE BUSINESS

Advancement of technology has changed the way businesses are conducted. The earlier system of brick and mortar has been replaced by the click and mortar and this is very different than the earlier one. Now, almost every type of business relies on computers for automating their traditional processes.

A large number of businesses are shifting to online platforms. Business by making online presence aims to reach to a global audience to increase their market share and other utilize this in a way to streamline their operations and costs. Businesses are largely dependent on computers and use them as a part of their daily operations along with payment processes and order management. There are many applications of e-commerce such as home banking, shopping in electronic malls, buying stocks, finding a job, conducting an auction, collaborating electronically with business partners around the globe, marketing & advertising and providing customer service. With the help of computers and technology online jobs can be done by sitting anywhere in case employees are not able to go outside. Now we can do our office work by sitting anywhere. Work From Home (WFH) , co working, job sharing, shared space are few of the popular examples in a philosophy of this direction.

2.6 COMPUTER ROLE IN ONLINE BANKING AND FINANCE

The emerging technologies of these days have transformed the banking industry from paper and pen to the digitized and networked services. Technology has revamped the accounting and management system of banking operations and it is now also focusing on changing the way how banks are delivering services to their customers. Along with these, computers have also transformed the financial industry and the way business deals are conducted from offline meeting to the online spaces. Accountants can now keep large amount of data to conduct complex calculations and manage financial transactions all from a computer than storing the data in books. Various applications of computers in online banking and finance are explained below:

2.6.1 Online Banking

- 1) **Plastic Money:** Popularity of credit cards or smart cards like Visa/Maestro has made the banking industry more attractive and flexible than ever before. By using online banking services, customers can very easily apply for smart cards or credit by sitting at their homes and the same can conveniently be delivered to them within a few days this do away the hassles of visiting the bank branches. Credit cards help the customers to borrow a specific amount from the bank for purchasing anything and to repay later. Smart card enables customers to instantly pay for anything and money is on the spot deducted from their bank accounts automatically. Smart cards can also be used for same card to depositing or withdrawing money from their accounts using an ATM machine.

- 2) **Self-Inquiry Facility:** Self enquiry facility provided by banks has actually reduced a lot burden of banks. Now instead of lining up or going to the help desks at the physical branches, customers can avail simple self-inquiry facility to get all the latest information by visiting their bank's website. This saves time on both the sides.
- 3) **Remote Banking:** Banks these days are providing various services online. Customers are no more required to travel for miles and stand in long waiting line to get their banking transactions done such as withdrawals, deposits and another enquiry. Customers now can install various banking app or login to banks website to get their transactions done as per their convenient. Remote banking has immensely help people in rural areas to save their hassles of coming to bank branches and to improve their culture of saving money.
- 4) **Signature Retrieval Facilities:** Computers by using internet technologies have provided services of digital authentication such as E-signature or digital signatures to safeguard the interest of its clients. By using this facility banks verifies customer's signatures while withdrawing a large sum of money from a specific bank account this reduce the chances of frauds, errors/risks that may arise due to forgery.
- 5) **Centralized Information Results to Quick Services:** Centralized storage of information with the banks enables them to transfer information from one branch to another effortlessly with not much time and costs involved and this results in quick services to customers. For example, if a customer registered his account with a rural branch, he can easily get details of his account at the main branch in an urban area.

2.6.2 Finance

- 1) **Financial Trading:** Computer based trading is useful when a trader has a system that allows profitable trading. Computer systems can even trade on the behalf of the users. A computer system can be programmed automatically to execute buy and sell orders when the price of a stock or bond reaches to a certain predefined level, and automatically closes the order when the target price reached below a certain level. Information technology gives instant information to stock traders for making decisions, and allowing them to enter orders that are immediately executed.
- 2) **Financial Information:** Through internet computers can have access to wide pool of financial information through. This enables business operators and individual's access to the information on investment

prospects that helps in conducting a detailed research on its profitability. Apart from this various other financial information such as world prices, tax changes, inflation rate and currency exchange rates are also available on internet which make easier for institutions and individuals to make informed financial decisions.

- 3) **Organizing and Analysing Data and Files:** Computers are very useful in organizing the data efficiently. This makes it easier for anyone to organize and find the information with a few clicks. Computers also help in the facilitation of data analysis. Employees can use various technologies available to understand complex spreadsheets. For example, Microsoft Excel can help businesses to visualize their data in charts and graphs so that trends and anomalies present in data can be spotted easily. This helps the businesses in making more informed decisions. Thus, enables organizations and individuals to plan finances without wasting much of time.
- 4) **Data Storage:** Keeping financial records is critical for organizations, as well as individuals, as it allows tracking of payment records, debts owed, purchases done and bank transactions made. Computers make data storage easily manageable and less bulky. Computers can also enable organizations to store large amounts of files in a small space that allows them to have a large track of historical transactions.
- 5) **Convenience:** Personal finance can be simplified by using information technology. Banks provide data on checking and savings deposits and withdrawals in standardized formats. Customer can easily download account transactions and store records on their computers. Personal finance software includes few additional features like charts and reports that show home users what they are spending money on and where their funds are coming from.
- 6) **Budgeting and Bookkeeping:** Computers are immensely helpful for companies considering financial transactions. Computer systems calculate and display the interest and principal of a loan, and estimate the returns on investment when the company borrows money to expand its operations. Companies can securely transfer data online, and the computer system records all transfers, which simplifies bookkeeping.

2.7 IMPORTANCE OF COMPUTER NETWORKS

Computer networks also known as data networks are chains of nodes linked by communication channels. These nodes can receive, transmit and exchange data between endpoints. The data transmitted can be in the form of voice or video traffic. Computer networks enable the users in accessing remote programs and remote databases either of the same organisation or from different enterprises or public sources. Computer networks provide

communication possibilities faster as compared to other facilities. With these optimal information and communication possibilities, computer networks may increase the organisational learning rate, which is considered as the only fundamental advantage in the prevailing competitive environment.

Types of Computer Networks

There are various types of computer networks. The critical difference in each type lies in the extent of geographical areas they serve and their core purpose. The various types of computer networks are explained below:

- 1) Personal Area Networks (PAN):** Personal Area Networks are interconnected technology devices that serve a single individual within one facility. It is reachable within a distance of 10 meters and is appropriate for small offices or residences. IT enabled devices include telephones, computers, video game and peripheral devices etc.
- 2) Local Area Network (LAN):** Local Area Networks are limited to a single building like an official establishment. It covers a defined area. It is useful for resource sharing including file servers, printers and data storage. LAN hardware such as ether net cables and hubs are affordable to acquire and maintain. Small LANs can efficiently manage two to three computers, and large LANs can host thousands of servers. Internet connectivity can either be wired or wireless. It is popular with many establishments because of its higher speed and lower set up costs.
- 3) Metropolitan Area Networks (MAN):** Metropolitan Area Network covers a much larger area as compared to Local Area Network. MAN includes configured computers covering a town, city or campus. On the basis of connections, MAN can connect either a single area or cross several miles. MAN is basically a series of several LANs linked to create a greater network.
- 4) Wide Area Networks (WAN):** Wide Area Networks cover a larger geographical area as compared to the Metropolitan Area Networks and can connect a country or even continents. The computers are interlinked through various cables, optical fibers or satellites. The users can access the network through modems. WANs basically comprised several smaller networks of LANs and MANs. The internet which consists of networks and gateways is an example of a public WAN.
- 5) Home Area Network (HAN):** Home Area Network is a computer network limited to a defined boundary like a house or home office. It is a type of IP based LAN which can either be wired or wireless. HAN is a broadband connection available to users using a wired or wireless modem. It consists of shared devices like faxes, printers, scanners etc.

Advantages of Computer Networks:

Following are the advantages of computers networks

- 1) **Overcoming Geographic Distances:** Computer networks popularly helps in overcoming the geographical limitation in a city or even across countries. Computers that need to communicate with other computers are often located in different parts of the world, and needs computer networks to pass information across long distances to overcome those geographical limitations.
- 2) **Allowing Information Sharing:** Computer networks allow sharing of information by allowing connected people or computers to access information on databases distributed in varying locations. There are databases hosted on several computers. People or computers may need to access the data in these databases. In the absence of computer networks, it would take a long time in selecting and transporting the required data from the database to the requiring users. With the help of computer networks, anyone can conveniently access the databases for retrieving the data.
- 3) **Supporting Distributed Processing:** Computer networks help to support processing distributed in many different locations. These different locations can either be physically close together inside a single computer room, or they can be separated by a long distance across mountains and oceans. Physical transportation of the intermediately processed data is unrealistic. Hence, a computer network becomes crucial in distributed processing. It allows timely passing of semi-processed data between computers located in distributed locations.

Check Your Progress B

- 1) What is Plastic Money?

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- 2) Differentiate between LAN and WAN.

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- 3) How can computers lead to more convenience while sorting huge financial data and information?

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4) What is the role of computers in online business?

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2.8 LET US SUM UP

For every kind of business, we can think of using computers in performing day to day operations in one way or another. The survival of these days' businesses is not possible without online presence. From generating reports to communicating with clients, business needs computers for performing almost each and every task. The uses of computer have brought away the business a long way ahead from the age of pen, paper, and their uses are simply countless. In the business organizations, computers help in many areas ranging from communication, marketing, and documentation to research. Technology is helpful in understanding behaviour of the users, how users interact in our site and what minor tweaks can be helpful in finding their right products.

Computers play an integral role in our day to day lives. Along with their uses at offices and home they are popularly being used in all kinds of sectors and businesses. They are used at airports, restaurants, railway stations, banks etc. Computers can also drastically change the way agricultural tasks and businesses being carried out across the world. In agriculture, computers are useful in finding out the best possible kinds of soil, plants and checking which match of these would result in the perfect crops.

The banking sector has popularised the use of technology in serving its client's firstly and also to do more with less. The emerging technologies of these days have transformed the banking industry from paper and pen to the digitized and networked services. Technology has revamped the accounting and management system of banking operations and it is now also focusing on changing the way how banks are delivering services to their customers.

Computer networks also known as data networks are chains of nodes linked by communication channels. These nodes can receive, transmit and exchange data between endpoints. The data transmitted can be in the form of voice or video traffic. Computer networks enable the users in accessing remote programs and remote databases either of the same organisation or from different enterprises or public sources.

2.9 KEY WORDS

Computer: Computer is an electronic device which helps in computations in which we feed input and get the results. Computers can vary on the basis of size, capacity and technology used.

Organization: An organization refers to a group of people working together for a common goal such as a corporation, a union, a neighborhood association or a charity.

Society: A society is a collective group of individuals who are involved in persistent social interaction or a large social group sharing the same territorial boundaries and are typically subject to the same political authority and dominant cultural outlooks.

Trade: Trade refers to a basic economic concept that involves the buying and selling of goods and services in return of compensation paid by a buyer to the seller.

Commerce: Commerce involves the exchange of goods and services on a large scale. It includes legal, economic, political, social, cultural and technological systems prevalent in a country or in international trade.

Computer Networks: Computer networks or data networks are chains of nodes linked by communication channels. Computer networks enable the users in accessing the remote programs and remote databases either of the same organization or from different enterprises or public sources.

2.10 TERMINAL QUESTIONS

- 1) What are the various applications of computers for organisations?
- 2) How are computers helping the society for betterment?
- 3) What do you mean by Wide Area Network? How does it differ from a local area network?
- 4) What is role of computers in online business?
- 5) What are the applications of computers for business, trade and commerce?
- 6) What are the Computer Networks? Explain their importance and various types.
- 7) Explain the utilities of computers in online Banking and Finance.

Note: These questions are helpful to understand this unit. Do efforts for writing the answer of these questions but do not send your answer to university. It is only for your practice.